

(EDL manuscript)
**Time-step-free Physics-Based Modeling of
Retention Loss in Cryogenic 3-D NAND Flash**

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진명

Detailed Poisson Solution / Q- ΔV_{th} Relation

Potential Solution: Poisson equation

$$\begin{cases} V_{Si}(r_{Si}) = V_{ch} \\ V_{TOX}(r) = V_{ch} + \frac{K_{TOX}}{\epsilon_{TOX}} \ln \frac{r}{r_{Si}} \\ V_{CTN,i}(r) = V_{CTN,i-1}(r_{i-1}) + \frac{K_i}{\epsilon_{CTN}} \ln \frac{r}{r_{i-1}} \\ V_{BOX}(r) = V_{CTN,N}(r_N) + \frac{K_{BOX}}{\epsilon_{BOX}} \ln \frac{r}{r_N} \\ V_{BOX}(r_{BOX}) = V_G \end{cases}$$

$$\mathbf{x} = [K_{TOX} \quad K_1 \quad \cdots \quad K_N \quad K_{BOX}]^T$$

Solution

$$A\mathbf{x} = \mathbf{b}$$

Boundary conditions

$$A = \begin{bmatrix} 1 & -1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & -1 & \cdots & 0 & 0 \\ \vdots & & & \ddots & & \vdots \\ 0 & 0 & 0 & \cdots & 1 & -1 \\ \frac{2\pi}{C_{TOX}} & \frac{2\pi}{C_1} & \frac{2\pi}{C_2} & \cdots & \frac{2\pi}{C_N} & \frac{2\pi}{C_{BOX}} \end{bmatrix}$$

$$\mathbf{b} = [\frac{\Delta Q_0}{2\pi} \quad \frac{\Delta Q_1}{2\pi} \quad \cdots \quad \frac{\Delta Q_N}{2\pi} \quad \Delta V]^T$$

$$\Delta Q_i = \pi r_i^2 (n_{t,i} - n_{t,i-1})$$

$$\Delta V = (V_G - V_{ch}) - \sum_{i=1}^N \frac{qn_{t,i}}{4\epsilon_{CTN}} (r_i^2 - r_{i-1}^2)$$

Q- ΔV_{th} Relation

>> computation optimization

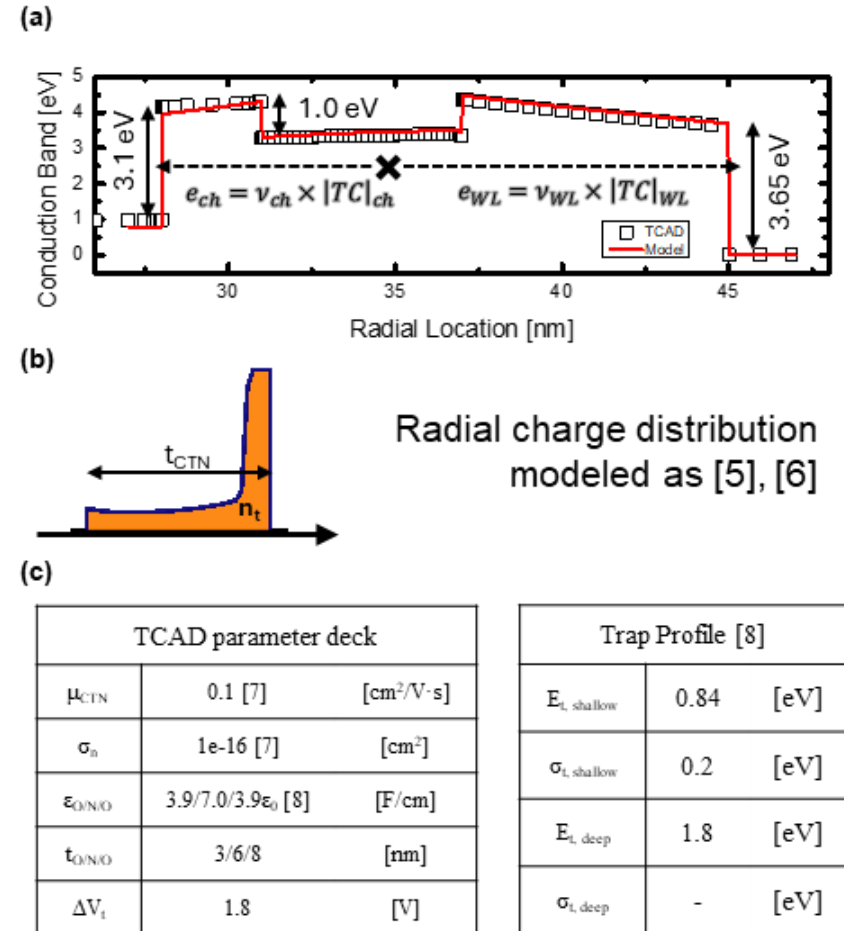
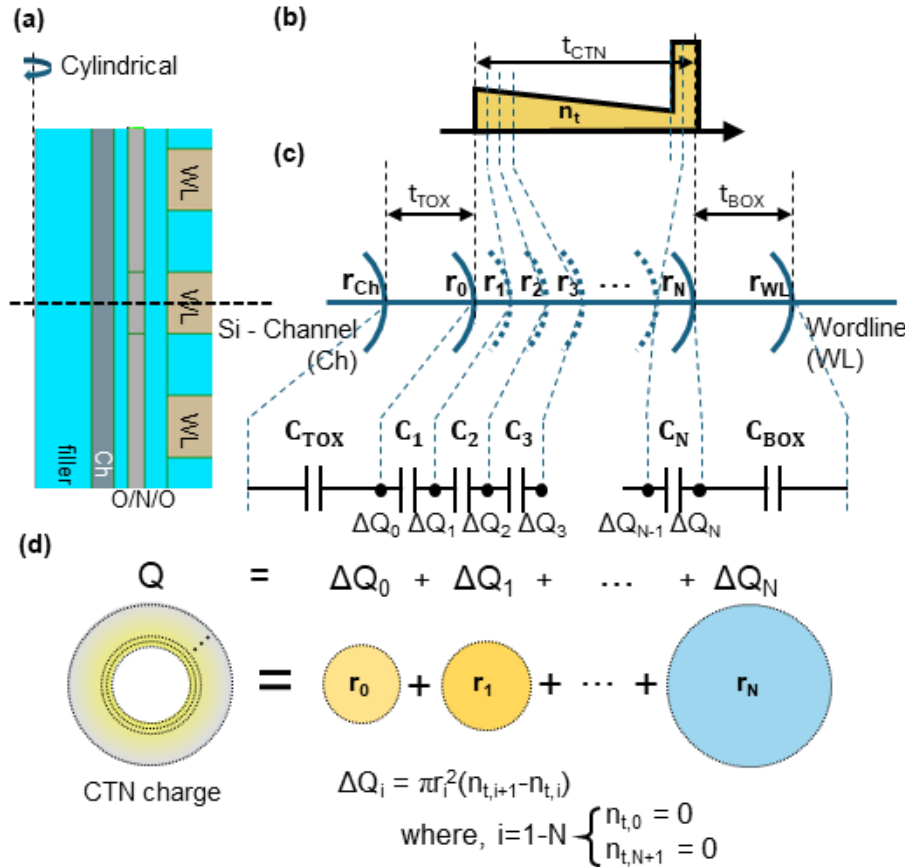
ΔV_{th} calculation can be done with
single determinant

$$K_{TOX} = \frac{\det [A_{\mathbf{b}(\Delta Q_i, \Delta V_{th})}]}{\det A}$$

$$\det [A_{\mathbf{b}(0, \Delta V_{th})}] = \det [A_{\mathbf{b}(\Delta Q_i, 0)}]$$

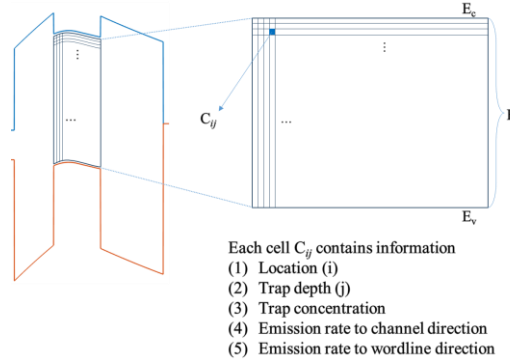
$$\therefore \Delta V_{th} = -\det [A_{\mathbf{b}(\Delta Q_i, 0)}]$$

Visualization & Parameters



Time-step-free Physics-based Modeling

Initial state definition



$$e = \nu \times |TC|$$

$$\nu = \frac{2\pi}{\hbar} |E_t|^2 V_{\text{trap}} \cdot \frac{m_0 \sqrt{2m^*}}{\pi^2 \hbar^3} \sqrt{E - E_{\text{trap}}}$$

$$|TC| = \prod \exp \left(-\frac{4\sqrt{2m^*}}{3\hbar a} \frac{\Phi_H^{1.5} - \Phi_L^{1.5}}{F} \right)$$

Emission evaluation
Jin, IEEE TED (2025)

$$\frac{d}{dt} n_t(r, E_t, t) = -(e_{\text{ch}} + e_{\text{WL}}) n_t(r, E_t, t) \quad (12)$$

Then, detrapping relation can be express as following autonomous ODE solution.

$$n_t(r, E_t, t) = \sum_r \sum_{E_t} n_{t0}(r, E_t) \exp(-(e_{\text{ch}} + e_{\text{WL}})t) \quad (13)$$

1. Initial state
 $n_{t0}(r, E_t)$

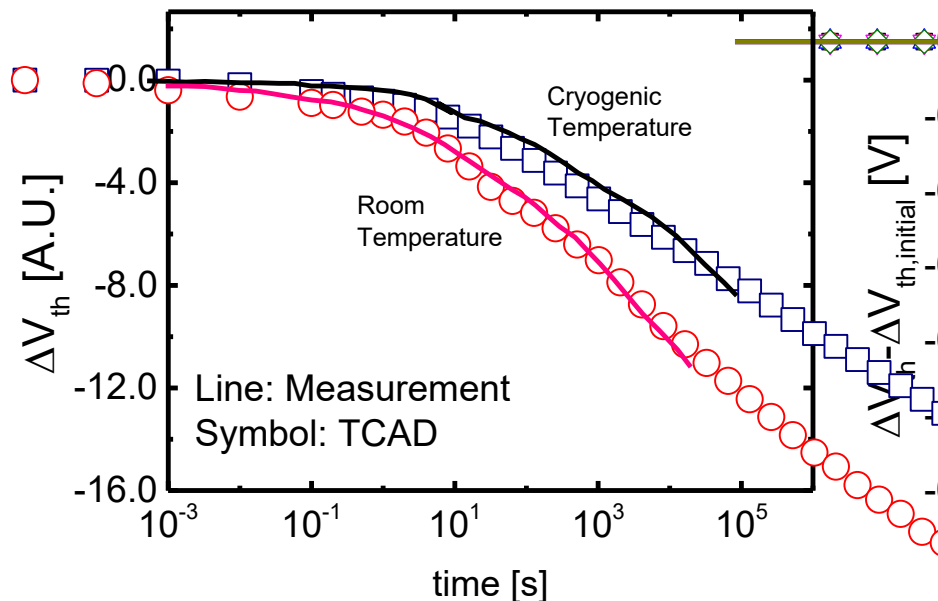
2. Emission rates
 $e_{\text{ch}}, e_{\text{WL}}$

3. Direct evaluation
 $n_t(r, E_t, t)$

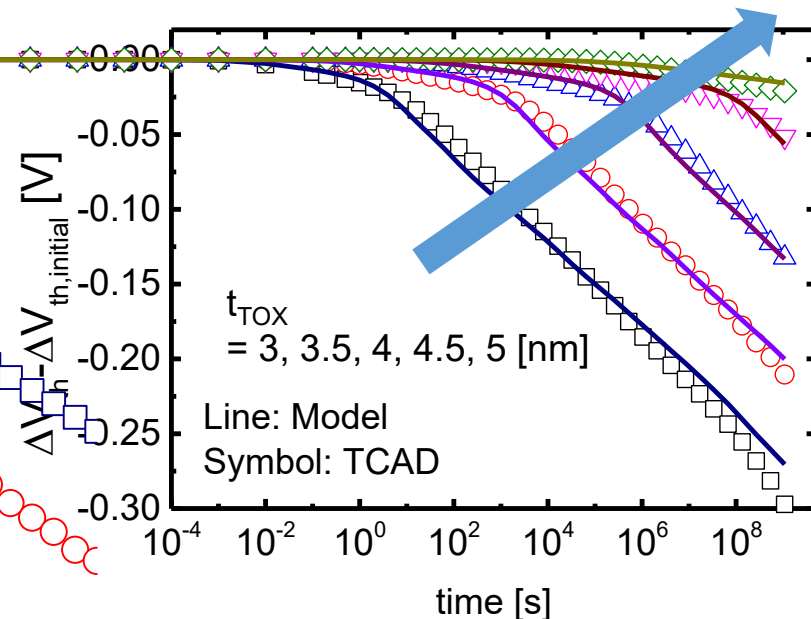
Cryogenic Calibration

Measurement (Refaldi, IRPS 2025)

Measurement – TCAD calibration



model – TCAD match



- TCAD cryogenic: $\sigma \rightarrow 0 \text{ cm}^2$
- O/N/O = 3/6/8

- $t_{TOX} = 3 \sim 5 \text{ [nm]}$

Computational Impact

Property	Transient TCAD	Time-step-free model
Temporal treatment	Sequential time stepping	Closed-form evaluation of Eq. (13)
Final-time dependence	Runtime grows as simulated retention time increases	Independent of final physical time for fixed queries
Average runtime	27483 s	3.926 s


≈ 7000×
 runtime reduction

- TCAD 는 final physical time까지 time step 별 순차적으로 계산
- Time-step-free model은 closed-form evaluation
- Computational resource는 final retention time 자체가 아니라 trap bin 수와 evaluated time point 수에 의해 결정